Meritocratic Hubris? (Over)confidence and Fairness Views

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Abstract

Feedback and evaluations shape modern life and are increasingly inflated across educational, professional, and digital systems. We hypothesize that inflated feedback, by boosting confidence, shifts fairness perceptions toward greater acceptance of inequality, even absent any material stake. In a preregistered online experiment with a US-representative sample, we subtly inflate performance feedback for identical test performances, which raises both overestimation and overplacement. Acting as impartial spectators, treated participants redistribute significantly less to lower-performing workers, reducing the extent of redistribution by about 14%. The effect is strongest among those who attribute performance to skill and effort, and text analysis points to motivated attention to performance as the operative channel.

1 Introduction

Feedback and evaluations shape modern life, from school grades and workplace promotions to algorithmic recommendations. Beyond signaling competence and guiding effort, feedback shapes how individuals perceive themselves and others. Crucially, feedback is increasingly inflated across educational and professional systems, as evidenced by widespread inflation of grades (Denning et al., 2022; Gans and Kominers, 2026) and performance evaluations (Bretz Jr et al., 1992; Bol, 2011; Golman and Bhatia, 2012) and flattering automated interactions with digital and AI platforms (Wei et al., 2023; Carro, 2024). AI chatbots engage in sycophancy in over 70% of interactions, providing

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positive affirmation in 36% of messages (Moore et al., 2026). This reflects a structural feature of how these systems are trained: reinforcement learning from human feedback (RLHF) optimizes for user approval rather than accuracy (Christiano et al., 2017; Sharma et al., 2023). Such inflated feedback may artificially boost individuals' confidence, their beliefs about their performance and abilities, which, via biased self-perception (Heidhues et al., 2018; Hestermann and Le Yaouanq, 2021), motivated reasoning (Bénabou and Tirole, 2016), or selective attention to performance (Lang et al., 2013; Loewenstein and Wojtowicz, 2023), could shift perceptions of fairness, potentially leading individuals to accept greater inequality.

This may have broad social implications. Economic inequality has sharply increased in advanced democracies (Chancel et al., 2022), yet public support for redistribution has stagnated or even declined (Kuziemko et al., 2015; Cappelen et al., 2007; Cavaillé, 2023). Higher education, often perceived as promoting egalitarian views, actually predicts lower support for redistribution (Bullock, 2021). Scholars attribute this to "meritocratic hubris" (Sandel, 2020), the tendency to see personal outcomes as deserved rewards for skill and effort rather than luck or structural advantages. We propose that meritocratic hubris emerges not only from genuine achievement but also from increasingly inflated signals of success, such as pervasive grade inflation or flattering evaluations by digital technologies like AI chatbots.

We hypothesize that inflated feedback, by boosting confidence, shifts individuals' fairness perceptions toward greater acceptance of inequality, even when they have no direct material interest at stake. Testing this hypothesis convincingly requires a design that disentangles confidence from other influences on fairness views. Observational data cannot do this, given confounding from reverse causality and unobserved individual characteristics. We therefore designed a preregistered online experiment with 800 participants representative of the US population. Using a minimal yet realistic intervention, we subtly inflated feedback by providing a lenient evaluation treatment for identical test performances. This isolates the causal effect of confidence (defined as beliefs about one's own performance) on fairness perceptions, while preserving ecological validity. The intervention increased both overestimation (beliefs about absolute performance) and overplacement (beliefs about relative performance). Participants then acted as impartial spectators: they redistributed money between two *others* based only on relative performance. Experimental evidence confirms that real-world AI feedback creates the same dynamic: AI assistance improves measured performance but inflates self-assessed performance by more, generating a net negative calibration (Fernandes et al., 2026; Bastani et al., 2025).

We find that even this subtle inflation of feedback significantly reduces support for redistribution. Despite having no personal financial stake, participants who received lenient evaluations consistently allocated fewer resources to lower-performing individuals, resulting in a decrease in the Extent of Redistribution of 14.4% (Figure 2), highlighting how minor shifts in confidence alone can measurably erode support for equality. The effect was strongest among those who attributed performance primarily to skill and effort rather than luck. Moreover, qualitative responses indicate that inflated feedback heightened the salience of performance differences, leading participants to justify inequality on meritocratic grounds explicitly. This points to subtle cognitive mechanisms, particularly motivated attention, as a likely channel linking inflated confidence to fairness views. These findings imply meritocratic hubris can stem not only from genuine success but also from artificially inflated evaluations. As inflated feedback spreads across education, workplaces, and digital interactions, such as AI-based chatbots (Moore et al., 2026; Sharma et al., 2023), subtle confidence boosts may aggregate into broader societal acceptance of inequality. Crucially, market forces likely exacerbate this issue: because positive feedback can be produced at negligible marginal cost and consumers tend to prefer reassuring evaluations (Möbius et al., 2022; Unknown, 2025), educational institutions and digital platforms face incentives to inflate performance signals. This creates a market-driven externality unlikely to be corrected by competition alone, underscoring the need for policymakers to address the overlooked societal costs of inflated feedback. This study advances three strands of literature. First, while previous research has studied feedback's effects on individual outcomes such as educational achievement and productivity (Murphy and Weinhardt, 2020; Tekleselassie et al., 2025), we document spillovers onto redistributive preferences, an externality largely overlooked by institutions that provide evaluations. Second, although overconfidence is widely recognized as a pervasive cognitive bias (Moore and Healy, 2008; Daniel and Hirshleifer, 2015; Malmendier and Tate, 2015), its broader social consequences remain underexplored. Existing experiments typically manipulate confidence using paradigms (e.g., "easy–hard" task designs) that simultaneously alter task difficulty and thus confound shifts in confidence with changes in perceived deservingness (Moore and Healy, 2008; Hansson and Sund, 2023). We introduce a simple and realistic intervention, subtle inflation of performance feedback, that consistently increases multiple dimensions of confidence (overestimation and overplacement) while maintaining truthful communication and without changing actual performance or task difficulty. This allows us to isolate confidence as a causal driver of fairness views and inequality acceptance. Finally, our research contributes to the growing literature on psychological and contextual determinants of distributive

preferences (Stantcheva, 2021; Cappelen et al., 2022, 2007, 2020; Almas et al., 2022). Complementing prior studies that explored how experiences of success and failure shape redistribution preferences (Deffains et al., 2016; Cassar and Klein, 2019; Espinosa et al., 2020; Fehr and Vollmann, 2022), we isolate confidence as a causal mechanism driving attitudes toward redistribution using a minimal, realistic intervention that mimics naturally occurring feedback inflation. In doing so, we extend recent experimental evidence that AI interactions inflate self-assessed performance beyond actual gains (Fernandes et al., 2026; Bastani et al., 2025) to a downstream outcome not previously documented: redistributive preferences.

2 Setting

We conducted a preregistered online experiment with 800 adult participants recruited from Prolific, an online platform shown to yield high-quality survey data (Peer et al., 2022). The sample was designed to be representative of the US population along age (approximately), sex, race, and political affiliation. In our main analysis, we focus on the sample that complies with all pre-registered comprehension checks, leaving 749 observations. As explained in Methods, participants completed a five-minute analytical task and were then randomly assigned to one of two grading schemes: a Lenient condition, which inflated performance feedback, or a Strict condition, which, relatively, did not (Extended Data Table 1). The grading scheme was the only aspect that varied across groups. After receiving feedback, participants make a redistribution decision as impartial spectators (Cappelen et al., 2020). They were informed that Worker One had outperformed Worker Two and received \$6 under a winner-take-all scheme. Participants could leave the allocation unchanged or redistribute any portion to the lower-performing worker.

3 Inflated Feedback Increases Confidence

Inflated feedback significantly increased participants' confidence (Figure 1). Participants in the Lenient condition overestimated their raw score by 0.74 points out of 10 (two-tailed t-test, t-statistic = 6.64; $P < 0.001$; n = 749), compared to a mean overestimation of –0.22 points in the Strict condition. They also believed that others scored higher on average (+0.53, two-tailed t-test, t-statistic = 4.61; $P < 0.001$; n = 749), relative to the Strict participants who attributed an average of 5.69 points to others. However, their perceived advantage over others also grew disproportionately, indicating increased overplacement. Within the study, treated participants overplaced themselves

by 0.22 points (two-tailed t-test, t-statistic = 1.81; P = 0.071; n = 749) compared to 0.08 in the Strict group. In more general terms, the treatment increased the share of participants who reported weakly better analytical skills (relative to the average American) by 6.7 percentage points (two-tailed t-test, t-statistic = 2.94; P = 0.003; n = 749), compared to a baseline of 85% in the Strict condition.

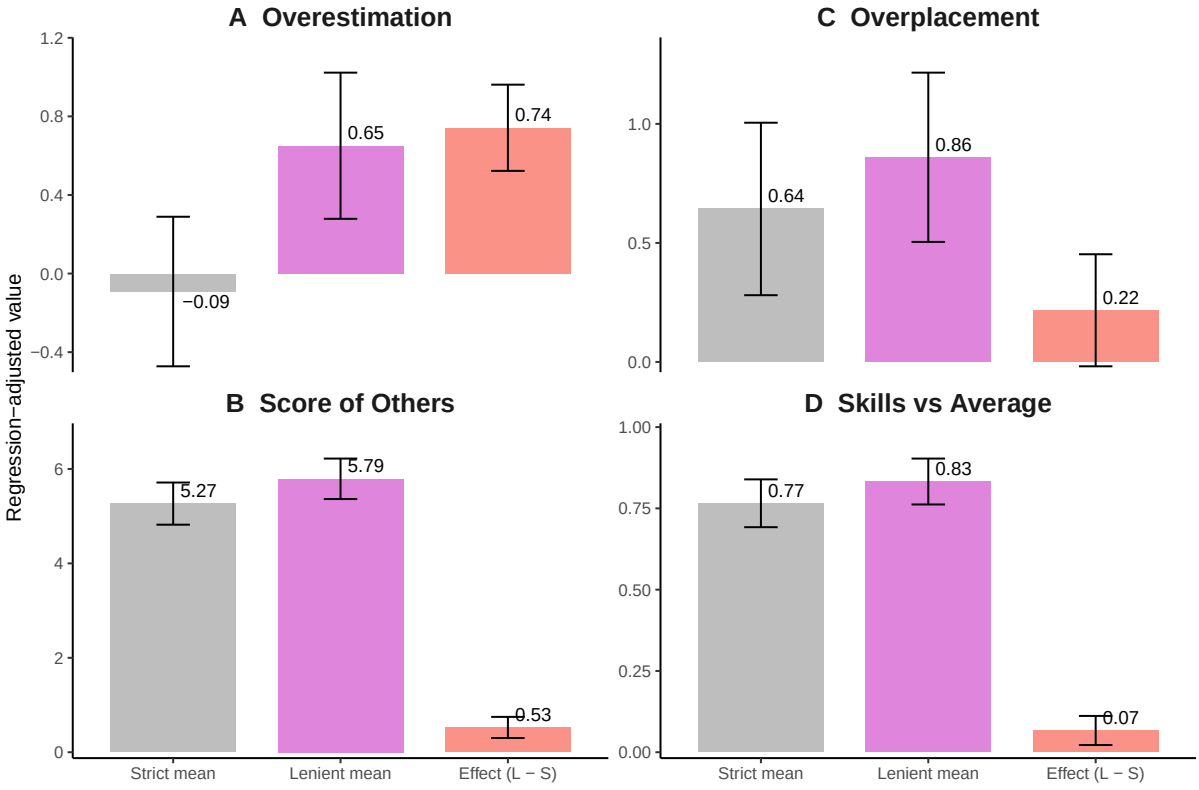


Figure 1: **Inflated feedback increases confidence measures.** Bars show regression-adjusted means (±95% confidence intervals) from OLS models including controls for task score, baseline confidence, and education level. The first two bars display adjusted means under strict and lenient evaluations, and the third bar shows the treatment effect (Lenient – Strict). Overestimation (Panel A) measures the gap between actual and perceived scores. Score of Others (Panel B) reflects the perceived scores of participants. Panel C captures the difference between perceived and actual score differences relative to the modal score. Panel D is a binary variable indicating whether individuals perceive their skills as weakly better than the average US skill level.

4 Confidence Reduces Support for Redistribution

The Lenient group allocated significantly more to the higher-performing worker (Figure 2). Relative to the Strict group, the Extent of Redistribution, as defined in Methods, fell by 0.05 (two-tailed t-test, t-statistic = –2.31, P = 0.021, n = 749), a 14.4% decline from the unadjusted Strict mean

of –0.34. Relaxing distributional assumptions, a Wilcoxon rank-sum test yields very similar results and remains statistically significant (Wilcoxon rank-sum test, P = 0.032, n = 749).

Lenient-treated individuals were significantly less willing to redistribute, suggesting their social preferences shifted toward less egalitarian outcomes.

Alternative outcomes (as in Methods) yield consistent results. Implemented Inequality rose by 0.05 (two-tailed t-test, t-statistic = 2.15, P = 0.032), and the Share Allocated to the Higher Performer Worker (Worker One) increased by 2.66 percentage points (two-tailed t-test, t-statistic = 2.31, P = 0.021) compared to the Strict mean of 66.8%. This effect is notable given the lack of personal financial stake: inflated feedback can shift fairness views, not just behavior driven by material self-interest.

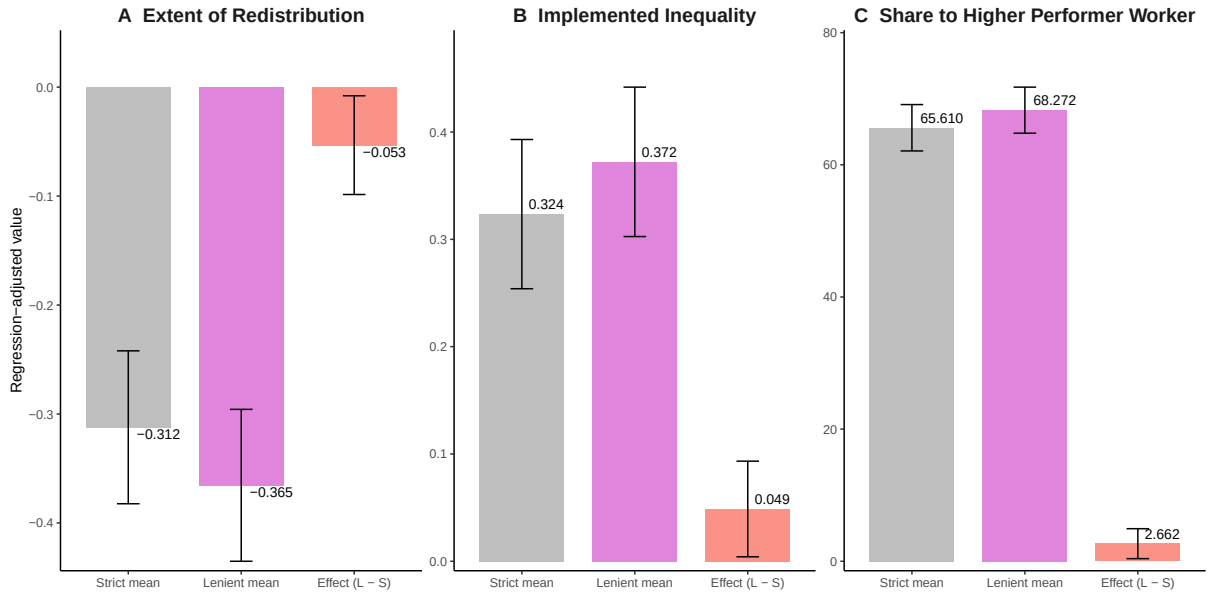


Figure 2: **Inflated feedback reduces support for redistribution.** Bars show regression-adjusted means (±95% confidence intervals) from OLS models including controls for task score, baseline confidence, and education level. The first two bars display adjusted means under strict and lenient evaluations, and the third bar shows the treatment effect (Lenient – Strict). The Extent of Redistribution (Panel A) is a measure between −1 and 1, where −1 allocates all to the higher performer (Worker One) and +1 allocates all to the lower performer (Worker Two). Implemented Inequality (Panel B) is the absolute counterpart measure and ranges from 0 to 1. Panel C, the Share Allocated to the Higher Performer, is the share given to the higher performer worker, plotted in percentage points.

5 Overplacement, Overestimation, and Redistribution

To identify which component of overconfidence drives redistributive preferences, we compare the effects associated with overestimation and overplacement. Conditional average treatment effect analyses (Athey and Imbens, 2015) (Extended Data Figure 1) point to overplacement as more predictive. Instrumental variable regressions confirm this pattern. First stage estimations validate the impact of inflated feedback on both measures of confidence, overestimation (two-tailed t-test, beta = 0.23, t-statistic = 7.20, P < 0.001, F-stat= 51.85, n = 749) and, in the right direction but weaker, overplacement (two-tailed t-test, beta = 0.07, t-statistic = 2.21, P = 0.028, F-stat= 4.87, n = 749). Second stage estimations show that overestimation is associated with a significant reduction in the support for redistribution (two-tailed t-test, t-statistic = –2.20, P = 0.028, n = 749), and overplacement points in the same direction although the estimate is noisier (two-tailed t-test, t-statistic = –1.62, P = 0.105, n = 749). The association with overplacement (–0.77 versus –0.23; Figure 3) tends to be larger but also noisier, presumably due to the weaker first stage. These results underpin the role of confidence and highlight relative standing as especially important for redistributive preferences, even when redistribution only affects others.

6 Heterogeneous Effects by Beliefs and Sociodemographics

Treatment effects varied across participants. The effect was strongest among participants who attributed performance to effort or skill rather than luck, supporting the idea that confidence reinforces beliefs that performance is merit-based (Figure 4). In addition, we observed larger treatment effects among younger participants and among those who identified as politically unaffiliated, consistent with feedback-induced confidence shaping fairness views most strongly among individuals with less entrenched ideological anchors (Ortoleva and Snowberg, 2015). In the age split, the treatment reduced redistribution significantly for participants aged younger than the median age of 42 in the US labor force (two-tailed t-test, beta = -0.09, t-statistic = -2.77; P = 0.0059, n = 327), whereas no effect was detected among those older (two-tailed t-test, beta = -0.02, t-statistic = -0.75; P = 0.45; n = 422). Similarly, the treatment had a measurable effect among independents (two-tailed t-test, beta = -0.12, t-statistic = -3.07; P = 0.0021; n = 276), while the corresponding effect among partisans was smaller and statistically indistinguishable from zero (two-tailed t-test, beta = -0.02, t-statistic = -0.55; P = 0.58; n = 473). Treatment effects were directionally consistent with the literature across gender and race, but none of the differences were statistically significant.

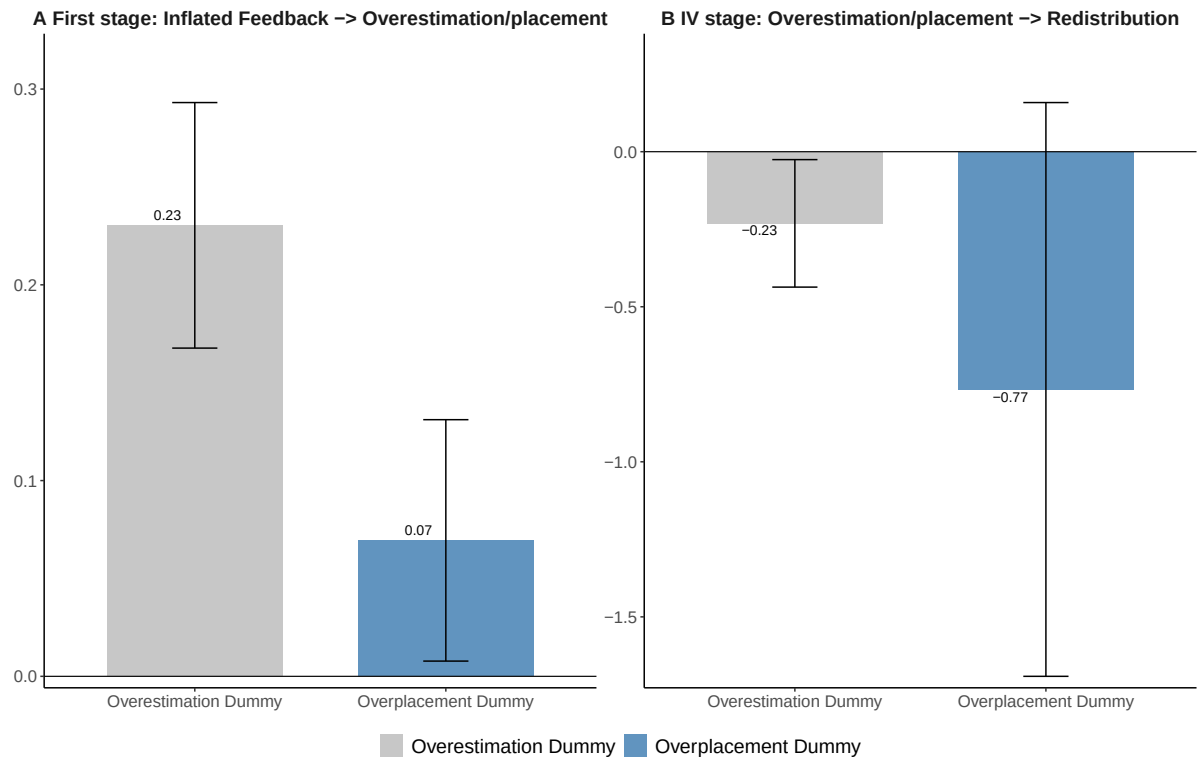


Figure 3: **Instrumental variable estimates of overestimation and overplacement.** Bars show regression-adjusted coefficients (±95% confidence intervals) from IV models including controls for task score, baseline confidence, and education level. Panel A reports first-stage regressions of inflated feedback on overestimation and overplacement. Panel B reports second-stage IV estimates of the effects of overestimation and overplacement on the Extent of Redistribution.

Nevertheless, we note that women reported lower baseline confidence than men, consistent with prior findings on gender gaps in confidence and competition (Extended Data Table 2) (Niederle and Vesterlund, 2007; Almås et al., 2020; Ranehill and Weber, 2022; Kuziemko et al., 2015; Alesina et al., 2018).

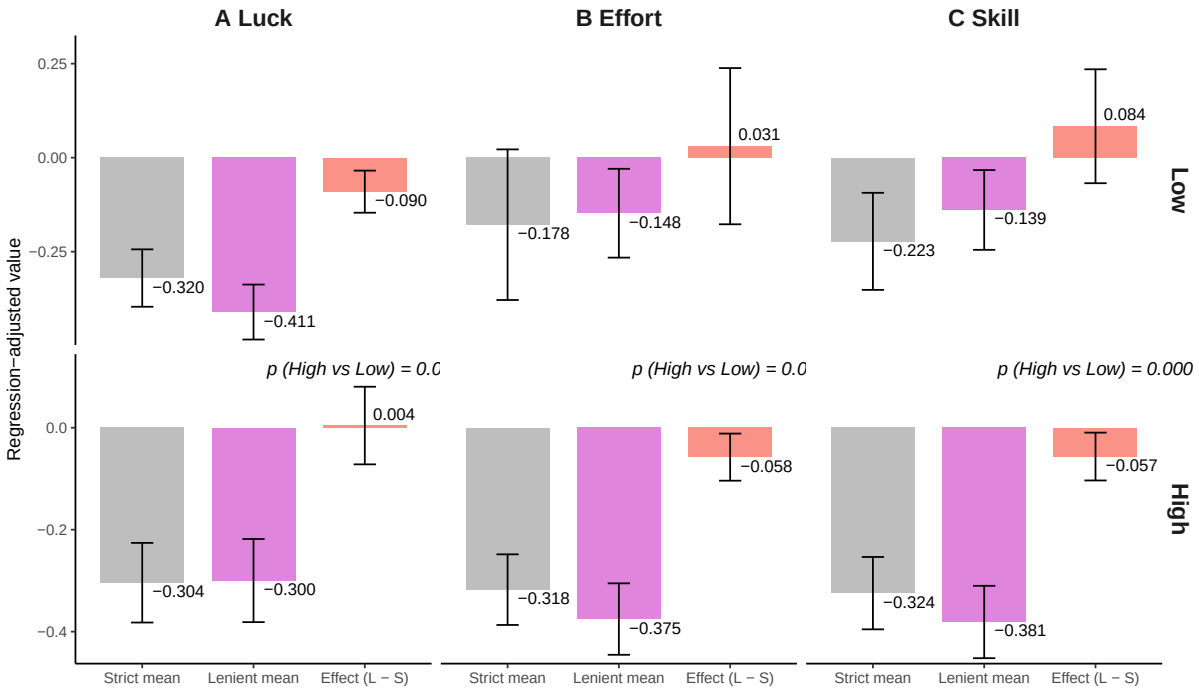


Figure 4: **Heterogeneity by perceived importance of luck, effort, and skill.** Bars show regression-adjusted means (±95% confidence intervals) from OLS models including controls for task score, baseline confidence, and education level. The first two bars display adjusted means under strict and lenient evaluations, and the third bar shows the treatment effect (Lenient – Strict) on the Extent of Redistribution. Panels are split by whether participants judged luck, effort, or skill to matter "not at all" or "a little" (*Low importance*) versus "somewhat," "quite a bit," or "very much" (*High importance*) for performance on the task. *p (High vs Low)* in High–importance panels reports the Welch t-test p-value comparing the treatment effect between High and Low groups (treated subsample).

7 Confidence and the Salience of Merit

To examine potential mechanisms, we analyzed participants' open-ended explanations of their redistribution choices. Exploratory text analysis indicates that inflated feedback made performance considerations more salient: performance-related words (e.g., "working," "answered," "right") were significantly more frequent in the Lenient condition, whereas redistribution-oriented terms (e.g.,

"distribution," "put") appeared more often in the Strict condition (Extended Data Table 3). To go beyond word counts, we classified each explanation along seven mutually non-exclusive categories of allocation justification using a large language model (GPT-5; see Methods). The merit category captures whether an explanation invokes Worker One's performance, effort, skill, score, speed, or task correctness as a relevant consideration in the allocation. In the Strict condition, 56.5% of explanations are classified as merit-based, compared to 63.0% in the Lenient condition, a difference of 6.5 percentage points (two-tailed z-test of proportions, $z = 1.81$; P = 0.069; n = 749; one-sided P = 0.035).

The same coding reveals a mirror-image substitution pattern. Appeals to strict egalitarian fairness, an even-split rationale grounded in equality rather than performance, fall from 23.5% in the Strict condition to 17.0% in the Lenient condition, a 6.5 percentage point decline (two-tailed z-test, $z = -2.19$; P = 0.028; n = 749). Lenient-treated participants thus move their explanations away from egalitarian framing and toward merit framing, even though the experimental manipulation provides no new information about the magnitude of the performance gap. Together, these patterns indicate that inflated feedback directs attention toward performance, subtly reshaping normative beliefs about fairness and deservingness in a manner consistent with motivated attention. By contrast, when explicitly measuring three previously pre-registered channels, self-serving attribution (Deffains et al., 2016), empathy (Andries et al., 2024), and affect, we found limited evidence supporting these mechanisms (Extended Data Table 4). Specifically, the treatment had no significant effect on beliefs about whether performance was driven by skill, effort, or luck, providing little support for self-serving attribution. Likewise, we observed no significant differences in empathy (measured as perceived similarity with the lower-performing worker) (Andries et al., 2024) or affect (measured via self-reported happiness). The divergence between explicit mechanism measures and open-ended responses aligns with recent evidence that attention, what comes spontaneously to mind, can shape behavior even in the absence of explicit belief updating (Haaland et al., 2024; Graeber et al., 2024) or new information (Bordalo et al., 2022, 2025). Our results, therefore, underscore the importance of capturing not only individuals' explicit beliefs, but also their implicit attentional focus when examining judgments of fairness and inequality.

8 Discussion

This study provides experimental evidence that inflated feedback reduces support for redistribution. A subtle manipulation, changing the grade attached to identical performance, was sufficient to increase participants' confidence and shift their fairness views. Treated individuals were significantly less likely to redistribute money to a lower-performing worker, despite having no personal stake in the outcome. These findings show that feedback affects not only self-perception but also distributive preferences. Because redistribution decreased even in an impartial setting, the move must reflect fairness views rather than self-interest. Moreover, the effect was stronger among those who attributed success to effort or skill.

Relative self-assessment, specifically overplacement, matters more than absolute overestimation. This highlights the role of social comparisons in shaping fairness views (Stantcheva, 2021). Contrary to some existing theories, we find little evidence that self-serving attribution, empathy, or affect mediate the results. Instead, exploratory text analyses point to attention as the operative channel, directing focus toward performance and subtly reshaping fairness judgments.

These results offer a psychological explanation for several macro-level puzzles. For instance, they may help explain why rising inequality has not been accompanied by a proportional rise in public support for redistribution (Cavaillé, 2023), and why higher education or career success is often associated with less egalitarian preferences (Bullock, 2021). When success or positive feedback breeds confidence, and confidence legitimizes inequality due to motivated attention to performance, support for redistribution decreases. The findings also raise practical questions. In educational or professional settings, grade inflation or overly positive feedback may have unexpected effects: it may shape how individuals think about fairness and merit. In digital environments, AI chatbots exhibit sycophancy in over 70% of interactions, with positive affirmation in 36% of messages (Moore et al., 2026). Experimental evidence confirms that this matters for self-assessment: AI assistance improves actual performance but inflates self-assessed performance by more, generating a net negative calibration (Fernandes et al., 2026), and AI-assisted students underperform on subsequent unassisted tests relative to those who never used AI (Bastani et al., 2025). Our results extend this chain: if AI-inflated confidence also reduces support for redistribution, the feedback structure of digital platforms may carry aggregate societal costs beyond individual learning outcomes.

9 Methods

9.1 Participants and Recruitment

We recruited 800 participants via Prolific, an online platform shown to yield high-quality survey data (Peer et al., 2022), between July 6 and August 13, 2024. The sample was stratified to approximate the joint distribution of age, gender, race, and political affiliation in the US population. Due to a platform constraint, the age filter was removed for the final five observations; the sample remains approximately representative in age. On average, participants were 46 years old, 50% female, 55% college-educated, and 63% employed at least part-time (Extended Data Table 5).

9.2 Payments and Incentives

Participants were paid at a base rate of $12 per hour. Additional performance-based incentives were provided.

Ten participants were randomly selected and paired. For these pairs, spectator redistribution decisions were implemented for real money.

Belief elicitation tasks, where participants predicted their own score and that of a random peer, were also incentivized. Five participants per question received $5 bonuses for correct predictions. For peer predictions, one participant was randomly selected as the benchmark against which guesses were evaluated.

We also incentivized a personality trait matching task, designed to assess empathy. Among the ten implemented redistribution pairs, one Worker Two per pair was randomly selected as the benchmark. All participants (including the selected Worker Two) had completed one of three short personality trait modules. For each module, we identified participants who exactly matched the benchmark's responses across five traits and correctly guessed their total score. From this group, two participants per module (six in total) received $5 bonuses.

9.3 Data Quality

We included several features to ensure high-quality responses. As highlighted in section Payments and Incentives, key responses were incentivized for accuracy (Danz et al., 2022).

Two strategically placed attention checks were used: one before the treatment assignment, the other before the mechanism question. Participants were also informed that completing the survey too quickly could lead to non-payment, following Prolific's policy.

Beyond formal checks, we aimed to foster engagement by framing participation as socially meaningful. We emphasized our role as non-partisan researchers and described the study's contribution to broader societal knowledge. We also highlighted participants' roles as important to the study's success.

Finally, participants were asked to reflect on how much attention they had paid to the survey, with the assurance that this would not affect their compensation.

Approximately 94% of participants passed all quality screens (sample restrictions are detailed in section Sample Definition and Statistical Analysis).

9.4 Experimental Procedure and Confidence Manipulation

The experiment consisted of five stages: (1) baseline confidence elicitation, (2) analytical task, (3) randomized feedback treatment, (4) redistribution decision, and (5) post-treatment belief and mechanism elicitation.

Participants first predicted the letter grade they expected to receive (A, B, or C), which we use as a measure of baseline confidence before receiving feedback.

They then completed a five-minute analytical reasoning task consisting of ten questions adapted from the Woodcock-Johnson cognitive test battery (Woodcock and Johnson, 1989). The number of tasks was not disclosed in advance, and the time constraint was tight, both of which increased subjective uncertainty about performance, key to allowing feedback to shift beliefs.

Participants were then randomized into one of two feedback conditions (Extended Data Table 1):

- The Strict (Control) condition mapped raw scores to conventional grade thresholds (e.g., 8/10 = B).
- The Lenient (Treatment) condition applied lower thresholds (e.g., 8/10 = A), mimicking grade inflation.

The only difference between conditions was the mapping from score to grade. Task difficulty, timing, and scoring were held constant. Randomization was implemented at the individual level using Qualtrics' built-in randomizer. This design addresses limitations of commonly used "easy–hard" paradigms, which shift confidence inconsistently: overplacement decreases, but overestimation often increases in task difficulty (Moore and Healy, 2008). In contrast, our Lenient treatment generates consistent shifts in both overestimation and overplacement without confounding changes in perceived effort or deservingness through changes in objective task difficulty.

The Lenient condition mimics real-world grade inflation, common in US higher education (Denning et al., 2022). More broadly, our Lenient treatment captures a general feature of many feedback environments, including algorithmic or workplace evaluations, where inflated or overly positive signals are common and can shape beliefs without altering actual performance.

9.5 Confidence and Beliefs Measures

Participants estimated their own and a peer's test score (0–10), allowing us to construct incentivized measures of overestimation and overplacement. Following Moore and Healy (Moore and Healy, 2008), overestimation was defined as:

$$\text{Overestimation} = \text{Perceived Own Score} - \text{Actual Own Score}$$

Overplacement was defined as:

$$\text{Overplacement} = (\text{Perceived Own Score} - \text{Perceived Peer Score}) - (\text{Actual Own Score} - \text{Modal Score})$$

where the perceived peer benchmark was incentivized to match the sample's modal score.

To capture broader confidence and perceived responsibility, participants reported whether they believed their analytical skills were above, below, or equal to the US average. They also rated the influence of skill, effort, and luck on their performance on a 5-point Likert scale.

Affect was measured using a 3-point Likert scale on overall happiness. To proxy empathy, participants completed a five-item personality inventory and guessed how many traits they shared with a lower-performing peer (0–5). This measure has been used as an indirect assessment of empathy, under the assumption that perceived similarity facilitates perspective-taking and concern for others (Andries et al., 2024).

Another measure of confidence is overprecision (Moore and Healy, 2008). It captures the belief that one's knowledge or estimates are more accurate than they really are. In principle, overprecision requires multiple measures of beliefs and a measure of a meta-belief about the accuracy of these beliefs. Given that we regard overprecision as a secondary measure of confidence for our purposes, we employ a simplified proxy to avoid substantial increases in cost and complexity. We elicit a range of scores that participants find plausible for themselves and proxy overprecision as:

$$\text{Overprecision} = (\text{Max Perceived Own Score} - \text{Min Perceived Own Score})$$

9.6 Outcome: Redistribution Decision

After receiving feedback, participants made a spectator redistribution decision; thus, redistribution choices only involved others. This is based on prior work on performance-based allocations that aims to capture fairness views (Konow, 2000; Cappelen et al., 2022). Participants were told that Worker One outperformed Worker Two on the same task and, under a winner-take-all rule, received \$6. Participants could leave the allocation unchanged or redistribute any portion to the lower performer.

Our main outcome is the Extent of Redistribution (Freyer and Günther, 2022), defined as:

$$d_i = \frac{I_i^{\text{Two}} - I_i^{\text{One}}}{I} \in [-1, 1]$$

where I_i^x is the amount spectator i allocates to Worker $x \in \{\text{One}, \text{Two}\}$, and $I = \$6$ is the total initial allocation given to the higher performer. Worker One is the higher earner before redistribution. This measure ranges from -1 (all to the higher performer) to $+1$ (all to the lower performer), with 0 representing an equal split. Pre-redistribution inequality is normalized to one, and complete equalization corresponds to zero post-redistribution inequality.

We also examine two secondary outcomes: Implemented Inequality (Almås et al., 2020), defined as

$$e_i = \frac{|I_i^{\text{Two}} - I_i^{\text{One}}|}{I} \in [0, 1],$$

and the Share Allocated to Higher Performer Worker (Worker One),

$$y_i = \frac{I_i^{\text{One}}}{I} \in [0, 1].$$

This impartial design avoids self-serving biases found in dictator games, allowing us to interpret redistribution as a normative fairness view rather than material self-interest.

9.7 Sample Definition and Statistical Analysis

We use a preregistered "attention sample" (N = 749) that excludes participants who failed both attention checks, reported low survey comprehension, indicated inattention, or gave open-ended responses inconsistent with the instructions. This sample is a subset of the 789 participants for whom complete data could be linked.

We estimate treatment effects (Lenient, $T = 1$) using ordinary least squares, with outcomes denoted by θ.

1. **Regression 1:**

$$\theta_i = \alpha + \beta T + \epsilon_i \tag{1}$$

2. **Regression 2:**

$$\theta_i = \alpha + \beta T + \delta \text{ score_FE} + \epsilon_i \tag{2}$$

   Here, we include the real score obtained in the task.

3. **Extended Regression:**

$$\theta_i = \alpha + \beta T + \gamma X_i + \delta \text{ score_FE} + \epsilon_i \tag{3}$$

   Baseline characteristics were generally balanced across experimental arms (Extended Data Table 5), with one exception: a small but statistically significant difference in college education. We include education as a control variable in X_i, as well as baseline confidence. Unless otherwise noted, inference uses heteroskedasticity-robust standard errors and t-based 95% confidence intervals.

9.8 Coding of Performance-Justified Inequality in Explanations

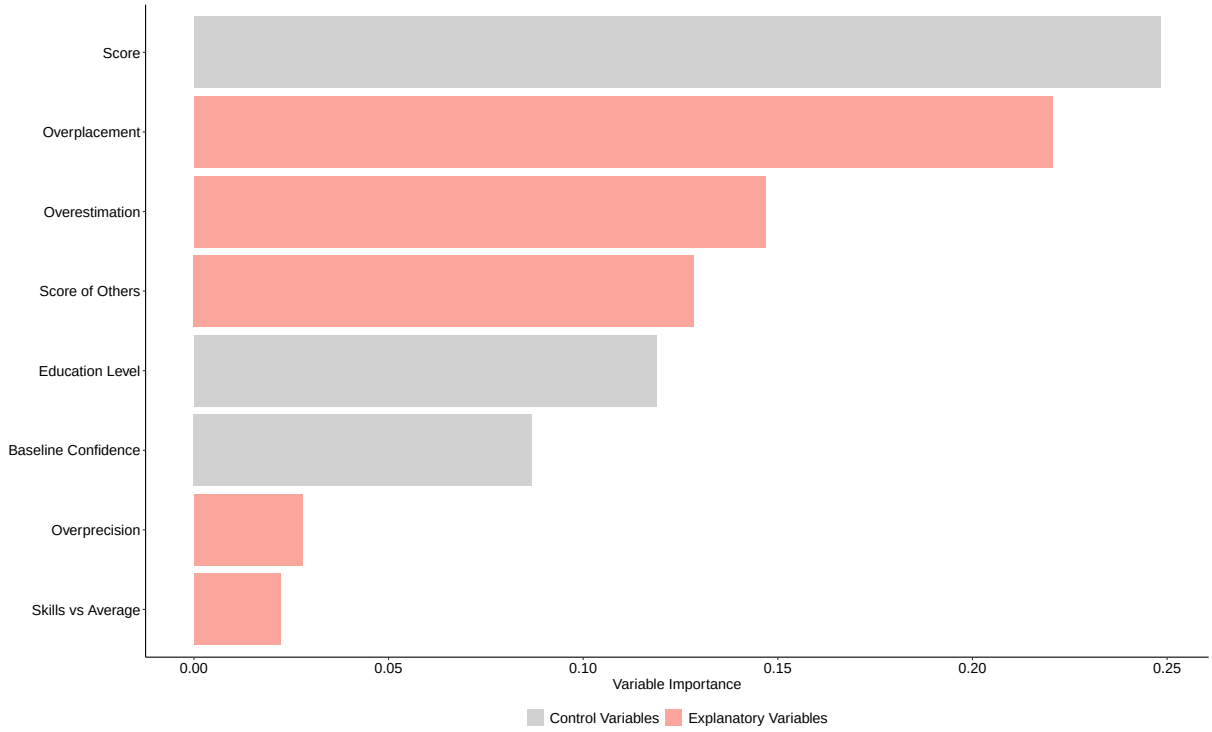
We classified each of the 749 open-ended explanations along seven mutually non-exclusive categories of allocation justification using a large language model.

We first prompted GPT-5, in a single API call with structured output, to propose between five and eight operationalizable categories spanning the space of justifications observed in the data. The model was given the paper's setting and central hypothesis but no prior coding scheme and no treatment labels. It returned seven categories: *merit_higher_performance* (invokes performance, effort, skill, score, speed, or task correctness as a relevant consideration in the alloca-

tion), *reward_effort_participation* (the lower performer deserves something for effort or participation), *equal_split_strict_fairness* (50/50 on equality grounds, independent of performance), *uncertainty_about_difference* (uses lack of information about the performance gap to justify the split), *respect_rules_or_contract* (invokes the stated rules or default allocation), *incentives_competition_logic* (frames the decision in terms of incentives, motivation, or opposition to participation prizes), and *equity_need_compassion* (appeals to compassion or non-merit equity).

We then applied GPT-5 to each of the 749 responses, requesting a binary 0/1 flag for each of the seven categories along with a one-sentence justification quoting the key phrase that drove each labeling decision. Each call used a fixed system prompt containing the category definitions and two positive plus two negative examples per category. The model received the response text and the participant's actual allocation amount to Worker One, but no treatment label. Calls were made independently and in parallel, with deterministic structured output via a Pydantic schema.

Our main indicator, *perf_inequality*, equals 1 when the LLM assigns the *merit_higher_performance* category and 0 otherwise. This operationalization matches our *a priori* definition: *perf_inequality* = 1 if the explanation justifies giving more money to Worker One by referring to Worker One's higher performance, effort, ability/skills, number or correctness of tasks, score, or speed; *perf_inequality* = 0 otherwise (including cases where such factors are mentioned only to be rejected as a basis for allocation). For example, statements such as "Worker 1 completed more tasks correctly so it is fair that they receive more of the bonus" are coded as *perf_inequality* = 1, whereas statements such as "Even though Worker 1 did more, I split the money equally because everyone should be treated the same" are coded as *perf_inequality* = 0.



Extended Data Figure 1 — Overplacement is more predictive than overestimation in conditional average treatment effects. Bars show variable importance scores from a causal forest predicting the conditional average treatment effect of inflated feedback on the Extent of Redistribution. Variables are grouped into *Explanatory variables* (salmon) and *Control variables* (grey). Among the Explanatory variables, *Overplacement* contributes most strongly to explaining heterogeneity in treatment effects, compared to overestimation and other belief measures.

References

Alesina, A., S. Stantcheva, and E. Teso (2018). Intergenerational mobility and preferences for redistribution. *American Economic Review 108*(2), 521–554.

Almas, I., A. W. Cappelen, E. Sorensen, and B. Tungodden (2022). Attitudes to inequality: Preferences and beliefs. *IFS Deaton Review of Inequalities*.

Almås, I., A. W. Cappelen, and B. Tungodden (2020). Cutthroat capitalism versus cuddly socialism: Are americans more meritocratic and efficiency-seeking than scandinavians? *Journal of Political Economy 128*(5), 1753–1788.

Andries, M., L. Bursztyn, T. Chaney, and M. Djourelova (2024). In their shoes. Technical report, National Bureau of Economic Research.

Score	Strict	Lenient
10	A	A
9	B	A
8	B	A
7	B	A
6	B	A
5	C	B
4	C	B
3	C	B
2	C	B
1	C	B
0	C	C

Extended Data Table 1 — Score-to-grade mapping under the two feedback conditions. The Strict condition uses conventional thresholds (e.g., a raw score of 8 maps to a B grade); the Lenient condition lowers the thresholds (e.g., 8 maps to an A). Task content, time limit, and scoring rule are held constant across conditions. Randomization is at the individual level.

Athey, S. and G. W. Imbens (2015). Machine learning methods for estimating heterogeneous causal effects. *stat 1050*(5), 1–26.

Bastani, H., O. Bastani, A. Sungu, H. Ge, O. Kabakci, and R. Mariman (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*.

Bénabou, R. and J. Tirole (2016). Mindful economics: The production, consumption, and value of beliefs. *Journal of Economic Perspectives 30*(3), 141–164.

Bol, J. C. (2011). The determinants and performance effects of managers' performance evaluation biases. *The Accounting Review 86*(5), 1549–1575.

Bordalo, P., J. Conlon, N. Gennaioli, S. Kwon, and A. Shleifer (2025). How people use statistics. *Review of Economic Studies*, rdaf022.

Bordalo, P., N. Gennaioli, and A. Shleifer (2022). Salience. *Annual Review of Economics 14*(1), 521–544.

Bretz Jr, R. D., G. T. Milkovich, and W. Read (1992). The current state of performance appraisal research and practice: Concerns, directions, and implications. *Journal of management 18*(2), 321–352.

	Gender		Race		Politics		Age	
	M	F	W	NW	Par	Ind	Younger	Older
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Lenient	-0.059^{*}	-0.051	-0.033	-0.087^{**}	-0.016	-0.120^{***}	-0.095^{***}	-0.023
	(0.033)	(0.034)	(0.029)	(0.038)	(0.029)	(0.038)	(0.034)	(0.031)
Mean Strict	-0.348	-0.326	-0.340	-0.330	-0.348	-0.319	-0.335	-0.337
P-value		0.6830		0.2660		0.0360		0.0467
Observations	373	376	470	279	473	276	327	422
R^2	0.073	0.072	0.072	0.090	0.043	0.126	0.105	0.042

Extended Data Table 2 — Heterogeneous treatment effects by demographics. Each column reports the coefficient on the Inflated Feedback Treatment from separate OLS regressions, estimated within the indicated subgroup and including controls for task score, baseline confidence, and education level. Robust standard errors are in parentheses; significance: * $p < 0.10$, $^{*}*$ $p < 0.05$, $^{*}**$ $p < 0.01$. *Mean Strict* reports the subgroup mean of the dependent variable in the control arm. The *P-value* row tests whether the treatment effect differs across the two levels of each split (e.g., Female versus Male) using a Welch t-test on treated–control differences. Definitions: *M/F* = Male/Female; *W/NW* = White/Non-White; *Par/Ind* = politically partisan/politically independent; *Younger* and *Older* = younger/older split compared to the median age of the US labor force (42 years old). Negative coefficients indicate reduced redistribution under lenient feedback.

Bullock, J. G. (2021). Education and attitudes toward redistribution in the united states. *British Journal of Political Science 51*(3), 1230–1250.

Cappelen, A. W., R. Falch, and B. Tungodden (2020). Fair and unfair income inequality. *Handbook of Labor, Human Resources and Population Economics*, 1–25.

Cappelen, A. W., A. D. Hole, E. Ø. Sørensen, and B. Tungodden (2007). The pluralism of fairness ideals: An experimental approach. *American Economic Review 97*(3), 818–827.

Cappelen, A. W., J. Mollerstrom, B.-A. Reme, and B. Tungodden (2022). A meritocratic origin of egalitarian behaviour. *The Economic Journal 132*(646), 2101–2117.

Carro, M. V. (2024). Flattering to deceive: The impact of sycophantic behavior on user trust in large language model. *arXiv preprint arXiv:2412.02802*.

Cassar, L. and A. H. Klein (2019). A matter of perspective: How failure shapes distributive preferences. *Management Science 65*(11), 5050–5064.

Word	Coefficient
working	1.10
extra	0.84
answers	0.67
distribution	-0.65
put	-0.61
questions	0.52
answered	0.49
right	0.40

Extended Data Table 3 — Distinctive words in redistribution justifications. This table reports words from open-ended justifications that are significantly associated with treatment assignment, ordered by the absolute magnitude of their coefficients (OLS regressions of treatment on TF–IDF weight, $p < 0.05$). Positive coefficients indicate words used more frequently by participants in Lenient, whereas negative coefficients indicate words used more frequently by participants in Strict.

Cavaillé, C. (2023). *Fair enough?: support for redistribution in the age of inequality*. Cambridge University Press.

Chancel, L., T. Piketty, E. Saez, and G. Zucman (2022). *World inequality report 2022*. Harvard University Press.

Christiano, P., J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei (2017). Deep reinforcement learning from human preferences. *Advances in Neural Information Processing Systems 30*.

Daniel, K. and D. Hirshleifer (2015). Overconfident investors, predictable returns, and excessive trading. *Journal of Economic Perspectives 29*(4), 61–88.

Danz, D., L. Vesterlund, and A. J. Wilson (2022). Belief elicitation and behavioral incentive compatibility. *American Economic Review 112*(9), 2851–2883.

Deffains, B., R. Espinosa, and C. Thöni (2016). Political self-serving bias and redistribution. *Journal of Public Economics 134*, 67–74.

Denning, J. T., E. R. Eide, K. J. Mumford, R. W. Patterson, and M. Warnick (2022). Why have college completion rates increased? *American Economic Journal: Applied Economics 14*(3), 1–29.

Espinosa, R., B. Deffains, and C. Thöni (2020). Debiasing preferences over redistribution: An experiment. *Social Choice and Welfare 55*(4), 823–843.

	Empathy	Affect	Self-Serving Attribution		
			Skills	Effort	Luck
	(1)	(2)	(3)	(4)	(5)
Lenient	0.004	0.017	0.023	0.002	0.063*
	(0.036)	(0.029)	(0.016)	(0.017)	(0.035)
Mean Strict	0.646	0.182	0.942	0.947	0.325
Observations	749	749	749	749	749
R^2	0.027	0.067	0.039	0.057	0.072

Extended Data Table 4 — Inflated feedback has little effect on preregistered mechanisms. The table reports coefficients from OLS regressions of empathy, affect, and self-serving attribution. Robust standard errors in parentheses; significance: * $p < 0.10$. Empathy is coded as 1 if individuals reported sharing at least 3 of 5 characteristics with a lower-performing peer. Affect is derived from a 3-point Likert scale on overall happiness and coded as a dummy for being "very happy" versus other responses. Self-serving attribution variables are coded as dummies equal to 1 if participants judged luck, effort, or skill to matter "somewhat," "quite a bit," or "very much" for performance, and 0 if judged "not at all" or "a little." Inflated feedback has no significant effect on empathy, affect, or attribution to skill and effort, and only a marginal effect on attribution to luck.

Fehr, D. and M. Vollmann (2022). Misperceiving economic success: Experimental evidence on meritocratic beliefs and inequality acceptance.

Fernandes, D., S. Villa, S. Nicholls, O. Haavisto, D. Buschek, A. Schmidt, T. Kosch, C. Shen, and R. Welsch (2026). AI makes you smarter but none the wiser: The disconnect between performance and metacognition. *Computers in Human Behavior 175*, 108779.

Freyer, T. and L. R. Günther (2022). Inherited inequality and the dilemma of meritocracy. Technical report, ECONtribute Discussion Paper.

Gans, J. S. and S. D. Kominers (2026, 5). What does a grade mean? informativeness and strategic manipulation of grading systems. NBER Working Paper 35183, National Bureau of Economic Research.

Golman, R. and S. Bhatia (2012). Performance evaluation inflation and compression. *Accounting, Organizations and Society 37*(8), 534–543.

Graeber, T., R. Hussam, K. Pandey, A. Shonchoy, and C. Yamauchi (2024). Lost in transmission. Technical report, HBS.

Variable	Strict Arm	Lenient Arm	Difference P-Value
Score	6.05 (2.39)	6.02 (2.48)	0.861
College Education Dummy	58.84 (49.28)	50.81 (50.06)	0.027
Income above Median Dummy	51.45 (50.05)	48.38 (50.04)	0.401
Non White	35.62 (47.95)	38.92 (48.82)	0.351
Age	45.55 (15.97)	45.84 (15.76)	0.800
Gender	53.03 (49.97)	47.30 (49.99)	0.117
Birth outside US	7.39 (26.19)	8.92 (28.54)	0.444
Non US nationality	0.26 (5.14)	0.27 (5.20)	0.986
First language not English	5.28 (22.39)	4.05 (19.75)	0.429
Student	12.14 (32.70)	15.41 (36.15)	0.195
Employed Full or Part Time	63.32 (48.26)	63.24 (48.28)	0.982
Republican	25.33 (43.55)	29.73 (45.77)	0.178
Democrat	29.29 (45.57)	30.54 (46.12)	0.709
Perceived Comprehension Survey	100.00 (0.00)	100.00 (0.00)	-
Perceived Unbiased Survey	94.46 (22.91)	94.32 (23.17)	0.936
Time to complete the set of tasks	4.58 (0.86)	4.56 (0.74)	0.651
Time to complete the survey	15.71 (6.29)	15.70 (6.25)	0.978

Extended Data Table 5 — Sample characteristics and baseline balance. This table reports the baseline characteristics across treatment arms. The average score out of 10 in the task was similar between the Strict (6.05) and Lenient groups (6.02), with no significant difference ($p = 0.861$). College education was significantly more common in the Strict group (58.84%) compared to the treatment group (50.81%) ($p = 0.027$). Other characteristics showed no statistically significant differences between groups at conventional levels.

Haaland, I., C. Roth, S. Stantcheva, and J. Wohlfart (2024). Measuring what is top of mind. Technical report, ECONtribute Discussion Paper.

Hansson, K. and O. Sund (2023). Lucky but confident—how confidence can polarize meritocratic beliefs and preferences for redistribution. *Available at SSRN 4527836*.

Heidhues, P., B. Kőszegi, and P. Strack (2018). Unrealistic expectations and misguided learning. *Econometrica 86*(4), 1159–1214.

Hestermann, N. and Y. Le Yaouanq (2021). Experimentation with self-serving attribution biases. *American Economic Journal: Microeconomics 13*(3), 198–237.

Konow, J. (2000). Fair shares: Accountability and cognitive dissonance in allocation decisions. *American economic review 90*(4), 1072–1092.

Kuziemko, I., M. I. Norton, E. Saez, and S. Stantcheva (2015). How elastic are preferences for redistribution? evidence from randomized survey experiments. *American Economic Review 105*(4), 1478–1508.

Lang, P. J., M. M. Bradley, and B. N. Cuthbert (2013). Motivated attention: Affect, activation, and action. In *Attention and orienting*, pp. 97–135. Psychology Press.

Loewenstein, G. and Z. Wojtowicz (2023). The economics of attention.

Malmendier, U. and G. Tate (2015). Behavioral ceos: The role of managerial overconfidence. *Journal of Economic Perspectives 29*(4), 37–60.

Möbius, M. M., M. Niederle, P. Niehaus, and T. S. Rosenblat (2022). Managing self-confidence: Theory and experimental evidence. *Management Science 68*(11), 7793–7817.

Moore, D. A. and P. J. Healy (2008). The trouble with overconfidence. *Psychological review 115*(2), 502.

Moore, J. et al. (2026). Characterizing and evaluating the reliability of llm sycophancy. *Working Paper*.

Murphy, R. and F. Weinhardt (2020). Top of the class: The importance of ordinal rank. *The Review of Economic Studies 87*(6), 2777–2826.

Niederle, M. and L. Vesterlund (2007). Do women shy away from competition? do men compete too much? *The quarterly journal of economics 122*(3), 1067–1101.

Ortoleva, P. and E. Snowberg (2015). Overconfidence in political behavior. *American Economic Review 105*(2), 504–535.

Peer, E., D. Rothschild, A. Gordon, and E. Damer (2022). Erratum to peer et al.(2021) data quality of platforms and panels for online behavioral research. *Behavior Research Methods 54*(5), 2618–2620.

Ranehill, E. and R. A. Weber (2022). Gender preference gaps and voting for redistribution. *Experimental Economics 25*(3), 845–875.

Sandel, M. J. (2020). *The tyranny of merit: What's become of the common good?* Penguin UK.

Sharma, M., M. Tong, T. Korbak, D. Duvenaud, A. Askell, S. Bowman, and E. Perez (2023). Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*. Published at ICLR 2024.

Stantcheva, S. (2021). Understanding tax policy: How do people reason? *The Quarterly Journal of Economics 136*(4), 2309–2369.

Tekleselassie, T., M. Witte, J. Radbruch, L. Hensel, and I. E. Isphording (2025). Feedback, confidence and job search behavior.

Unknown, A. (2025, Aug). Openai's rocky gpt-5 rollout shows struggle to remain undisputed ai leader. *The Wall Street Journal*. Online; accessed 2025-09-16.

Wei, J., D. Huang, Y. Lu, D. Zhou, and Q. V. Le (2023). Simple synthetic data reduces sycophancy in large language models. *arXiv preprint arXiv:2308.03958*.

Woodcock, R. W. and M. B. Johnson (1989). *Woodcock-Johnson tests of cognitive ability*. DLM Teaching Resources.